

Point Stabilizers in $\text{Aut}(\mathbb{Q}, <)$ Are Inj-Key but Not Co-Minimal

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Source problem

The source is Michael Megrelishvili, *Key subgroups in the Polish group of all automorphisms of the rational circle*, arXiv:2410.17905v3 [1]. In Section 4 (PDF page 16), after proving the analogous theorem for the rational circle, the author writes:

The Polish group $G := \text{Aut}(\mathbb{Q}, \leq)$. The linearly ordered rational case suggests a closely related example. It is expected, and the same method should establish, that for every $a \in \mathbb{Q}$, the stabilizer $H := G_a$ is inj-key but not co-minimal in the Polish group $G := \text{Aut}(\mathbb{Q}, \leq)$. The relevant greatest G -compactification $\beta_G(G/G_a)$ is a linearly ordered metrizable G -flow; it can be described as the compact interval $[0, 1]$ in which every internal rational point $q \in \mathbb{Q} \cap (0, 1)$ is replaced by an ordered triple q^-, q, q^+ . See [25, Remark 3.8] and also [26, Example 5.4(b)].

This G -compactification $\beta_G(G/G_a)$ has six G -orbits: the middle rational orbit $\mathbb{Q} \cap (0, 1)$, the two twin rational orbits $(\mathbb{Q} \cap (0, 1))^-$ and $(\mathbb{Q} \cap (0, 1))^+$, the irrational orbit $J = [0, 1] \setminus \mathbb{Q}$, and the two endpoint orbits $\{0\}$ and $\{1\}$.

Thus the focused assertion is that, for every $a \in \mathbb{Q}$, the point stabilizer in the Polish group $\text{Aut}(\mathbb{Q}, <)$ is inj-key but not co-minimal. We prove that assertion.

Definitions

Let (G, τ_0) be a Hausdorff topological group. Write $\mathcal{T}_\downarrow(G, \tau_0)$ for the Hausdorff group topologies on the abstract group G which are coarser than τ_0 . A subgroup $H \leq G$ is *inj-key* if the restriction map

$$\mathcal{T}_\downarrow(G, \tau_0) \longrightarrow \mathcal{T}_\downarrow(H, \tau_0|_H), \quad \sigma \longmapsto \sigma|_H,$$

is injective. It is *co-minimal* if $\sigma/H = \tau_0/H$ for every $\sigma \in \mathcal{T}_\downarrow(G, \tau_0)$.

Fix a countable dense copy $D = \mathbb{Q} \cap (0, 1)$ and put $G = \text{Aut}(D, <)$, endowed with the pointwise topology τ_0 obtained by making D discrete. Every $g \in G$ extends uniquely to an increasing homeomorphism $\bar{g}$ of $I = [0, 1]$. Denote by τ_I the topology induced from the compact-open topology of $\text{Homeo}_+(I)$.

We use three standard facts. First, τ_I is the minimum Hausdorff group topology on the abstract group G . This follows from the local-movement criterion of Chang–Gartside [2, Theorem 8]; the same criterion is used for the rational-circle group in the source. Second, Merson’s lemma says that if $\gamma_1 \subseteq \gamma$ are group topologies with

$$\gamma_1|_H = \gamma|_H, \quad \gamma_1/H = \gamma/H,$$

then $\gamma_1 = \gamma$ [5]. Third, the greatest equivariant compactification of the discrete G -space $G/G_a \cong D$ is the ordered compactum obtained from $[0, 1]$ by replacing every $q \in D$ by an ordered triple $q^- < q < q^+$ [3, 4]. It has the six orbits listed in the source crop.

Idea of the proof

There are two natural ways that a coarser Hausdorff group topology can see the orbit G/G_a : either every rational point is isolated, or the orbit has its ordinary order topology inherited from $[0, 1]$. The six-orbit compactification shows that there is no third possibility.

The only dangerous situation is therefore a topology which is still the full pointwise topology on G_a but sees G/G_a with the nondiscrete order topology. A small group neighbourhood for such a topology contains a finite pointwise stabilizer. Order ultrahomogeneity then gives a four-factor decomposition for every automorphism supported in a short interval. We use one such local automorphism to correct a small element back into G_a while preserving its motion of a second rational point. This contradicts the assumed pointwise topology on G_a .

The two quotient topologies

Lemma 1 (Coset-topology dichotomy). *Fix $a \in D$ and $H = G_a$. For every $\sigma \in \mathcal{T}_\downarrow(G, \tau_0)$,*

$$\sigma/H \in \{\tau_0/H, \tau_I/H\}.$$

Moreover, if $\sigma/H = \tau_0/H$, then $\sigma = \tau_0$.

Proof. Because τ_I is minimum and H is closed in τ_I , the subgroup H is closed for every $\sigma \in \mathcal{T}_\downarrow(G, \tau_0)$. Moreover, $\bigcap_{g \in G} gHg^{-1} = \{e\}$, so the action on G/H is effective. The general coset-compactification lemma used in the source therefore applies: every topology σ/H is realized by the compact-open topology of an injective G -factor of the greatest G -compactification of D .

We record the factor check in this linearly ordered setting. In the triple interval

$$B = \{0, 1\} \cup D^- \cup D \cup D^+ \cup J, \quad J = (0, 1) \setminus D,$$

the stabilizers of q^-, q, q^+ coincide with G_q . The actions on both D and the cut orbit J are order-2-transitive (piece together order isomorphisms on the three complementary rational intervals), hence primitive. Their point stabilizers are therefore maximal, so the restriction of a G -factor to each nontrivial orbit is either injective or constant. Notice also that each of D^-, D^+ , and J is dense in $B \setminus D$: the only points missed by its closure are the isolated middle points q . Consequently, if one of these three orbits is collapsed, continuity collapses all of $B \setminus D$. Each surviving middle point q is then clopen, so G_q is open in the induced compact-open group topology.

Suppose none of these three orbits is collapsed. If two points in distinct nontrivial orbits have the same image, injectivity on their respective orbits shows that their stabilizers are equal. The only such equalities are

$$G_{q^-} = G_q = G_{q^+} \quad (q \in D).$$

Thus, apart from a possible identification of the two globally fixed endpoints, the factor relation is obtained by applying uniformly one of the five partitions of the three labels q^-, q, q^+ . If the middle point is not identified with either twin, its singleton image is clopen. If $q^- \sim q$ but q^+ remains separate, then for $q < r$ the interval from the image of q^+ to the image of r^- is clopen; its setwise stabilizer fixes its endpoints and is contained in G_q . The case $q \sim q^+$ is symmetric. The remaining

non-total partition, $q^- \sim q^+$ with q separate, again leaves the singleton image of q clopen. In all four non-total cases some open subgroup is contained in G_q , so G_q itself is open. Since all conjugate point stabilizers are then open, their finite intersections form a neighbourhood base for τ_0 , and the induced topology is τ_0 .

The only remaining factor identifies $q^- \sim q \sim q^+$ for every $q \in D$. On the dense union of the rational and cut orbits this is exactly the canonical collapse $B \rightarrow [0, 1]$ (possibly followed by identification of the two globally fixed endpoints). Its induced topology on G is τ_I , and the orbit topology is the usual nondiscrete order topology τ_I/H . This proves the dichotomy.

Finally, if σ/H is discrete, then H is σ -open. Every conjugate G_q is σ -open, as is every finite intersection of point stabilizers. Those intersections form a base for τ_0 , whence $\tau_0 \subseteq \sigma \subseteq \tau_0$. $\square$

A localized factorization

For a finite chain $x_1 < \dots < x_n$ write $F = G_{x_1, \dots, x_n}$. For $3 \leq k \leq n-2$ set

$$M_k = \{f \in G : f(x) = x \text{ for every } x \notin (x_{k-1}, x_{k+1})\}.$$

Lemma 2 (Local W^4 lemma). *Let $\sigma \in \mathcal{T}_\downarrow(G, \tau_0)$ with $\sigma \neq \tau_0$. Let U be a symmetric σ -neighbourhood and suppose $F \subseteq U$. If the chain has at least three marked points on either side of x_k , then $M_k \subseteq U^4$.*

Proof. Fix $f \in M_k$. If $f(x_k) = x_k$, then $f \in F \subseteq U$. By symmetry we may suppose $f(x_k) > x_k$. Intersect U with a symmetric τ_I -neighbourhood V so small that, for every $v \in V$,

$$v(x_i) \in (x_{i-1}, x_{i+1}) \quad (2 \leq i \leq n-1), \quad v(f(x_k)) \in (x_k, x_{k+1}).$$

Put $W = U \cap V$. Since every element of F fixes the relevant interval endpoints, $F \subseteq W$.

There is $g \in W$ moving x_k ; otherwise G_{x_k} would contain a σ -neighbourhood, hence would be σ -open, forcing $\sigma = \tau_0$. Replacing g by g^{-1} if necessary gives

$$x_{k-1} < g(x_k) < x_k < f(x_k) < x_{k+1}.$$

The smallness of V implies that the open interval

$$(s, t), \quad s = \max\{x_{k-1}, g^{-1}(x_{k-1})\}, \quad t = \min\{x_{k+1}, g^{-1}(x_{k+1})\},$$

contains $g^{-1}(x_k)$ and $g^{-1}f(x_k)$ but contains none of $x_i, g^{-1}(x_i)$ for $i \neq k$. (The two extra marked points on each side ensure these inequalities after applying $g^{\pm 1}$.)

By ultrahomogeneity of the rational order inside (s, t) , choose $\varphi \in G$, supported in (s, t) , such that

$$\varphi(g^{-1}f(x_k)) = g^{-1}(x_k), \quad \varphi(x_k) = x_k.$$

The choice of (s, t) shows that φ fixes every x_i , so $\varphi \in F \subseteq W$. It also shows that $r = g\varphi g^{-1}f$ fixes every x_i , hence $r \in F \subseteq W$. Therefore

$$f = g\varphi^{-1}g^{-1}r \in W^4 \subseteq U^4.$$

The case $f(x_k) < x_k$ is the order-reversed version of the same argument. $\square$

Lemma 3 (The mixed case is impossible). *There is no $\sigma \in \mathcal{T}_\downarrow(G, \tau_0)$ such that*

$$\sigma|_H = \tau_0|_H, \quad \sigma/H = \tau_I/H.$$

Proof. Assume otherwise and choose $b \neq a$. Since the restriction to H is the pointwise topology, there is a symmetric σ -neighbourhood U with

$$U^5 \cap H \subseteq H \cap G_b. \quad (1)$$

The quotient hypothesis and Lemma 1 give $\sigma \neq \tau_0$. Choose a long finite chain $x_1 < \dots < x_n$ whose pointwise stabilizer F lies in U , and refine it so that $a \in (x_k, x_{k+1})$, while b is one of the marked points outside the closed interval $[x_{k-1}, x_{k+2}]$, with three marked points available on each side of x_k .

Because $\tau_I \subseteq \sigma$, intersect U with a sufficiently small τ_I -neighbourhood and call the result W . We may arrange that $F \subseteq W$, $W = W^{-1}$, and

$$w(a) \in (x_k, x_{k+1}), \quad w(b) \notin (x_k, x_{k+1}) \quad (w \in W). \quad (2)$$

Since $\sigma \neq \tau_0$, the subgroup G_b is not σ -open, so choose $w \in W$ with $w(b) \neq b$. Equation (1) shows that $w(a) \neq a$: otherwise $w \in W \cap H \subseteq U^5 \cap H \subseteq G_b$.

By (2) and order ultrahomogeneity in (x_k, x_{k+1}) , there is an $m \in G$ supported in this interval with $m(w(a)) = a$. It fixes $w(b)$, so $mw \in H$ and $(mw)(b) = w(b) \neq b$. On the other hand, $m \in M_k$, and Lemma 2 gives $m \in W^4$. Hence

$$mw \in W^5 \cap H \subseteq U^5 \cap H \subseteq G_b,$$

contradicting $(mw)(b) \neq b$. □

Main result

Theorem 1. *For every $a \in \mathbb{Q}$, the point stabilizer $H = G_a$ is inj-key but not co-minimal in the Polish group $(G, \tau_0) = \text{Aut}(\mathbb{Q}, <)$.*

Proof. First, H is not co-minimal: the topology τ_I is Hausdorff and coarser than τ_0 , but τ_I/H is the nondiscrete order topology on D , whereas τ_0/H is discrete.

For inj-keyness, take $\sigma_1, \sigma_2 \in \mathcal{T}_\downarrow(G, \tau_0)$ with $\sigma_1|_H = \sigma_2|_H$. By Lemma 1, each quotient is either τ_0/H or τ_I/H .

If both are τ_0/H , Lemma 1 gives $\sigma_1 = \sigma_2 = \tau_0$. If exactly one is τ_0/H , that topology is τ_0 ; the common restriction is therefore $\tau_0|_H$, contradicting Lemma 3 for the other topology.

It remains to treat the case $\sigma_1/H = \sigma_2/H = \tau_I/H$. Let $\sigma = \sup\{\sigma_1, \sigma_2\}$. The equality of the restrictions gives

$$\sigma|_H = \sigma_1|_H = \sigma_2|_H.$$

If $\sigma/H = \tau_I/H$, Merson's lemma applied to $\sigma_i \subseteq \sigma$ yields $\sigma_i = \sigma$ for $i = 1, 2$. If instead $\sigma/H = \tau_0/H$, Lemma 1 gives $\sigma = \tau_0$. Then $\sigma_i|_H = \tau_0|_H$ while $\sigma_i/H = \tau_I/H$, contradicting Lemma 3. Thus only the first alternative occurs and $\sigma_1 = \sigma_2$. □

Verification notes and limitations

All group topologies in the proof are Hausdorff. The use of the minimum topology guarantees that H remains closed, so every quotient is Hausdorff. The local factorization is purely finite and uses only order ultrahomogeneity; no completeness or Baire-category step is hidden in it.

The theorem solves only the explicit linearly ordered rational case proposed in Section 4 of the source. It does not answer Question 3.19 about examples where the canonical extremely amenable subgroup is not inj-key, nor the Menger-continuum and two-cell questions later on the same page.

Novelty check and human review

On July 22, 2026, local indexes and the source corpus were searched for arXiv:2410.17905, $\text{Aut}(\mathbb{Q}, <)$, `inj-key`, `co-minimal`, and point stabilizers. Web searches used the exact and near-exact phrases “ $\text{Aut}(\mathbb{Q}; i)$ inj-key point stabilizer”, “ $\text{Aut}(\mathbb{Q}; i=)$ co-minimal stabilizer”, and “linearly ordered rational inj-key subgroup”. They found the source and the earlier rational-circle result, but no proof of the linearly ordered case. The downloaded v3 source still labels the result as expected rather than proved.

Human review should focus first on Lemma 1: verify the orbit-factor check against the six-orbit triple-interval compactification. The second focus is the interval bookkeeping in Lemma 2. Once those two lemmas are accepted, the mixed-case contradiction and the final Merson argument are formal.

References

- [1] M. Megrelishvili, *Key subgroups in the Polish group of all automorphisms of the rational circle*, arXiv:2410.17905v3, 2026.
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- [5] D. Dikranjan, I. Prodanov, and L. Stoyanov, *Topological Groups: Characters, Dualities and Minimal Group Topologies*, Pure and Applied Mathematics 130, Marcel Dekker, 1989.